MYSQL MINI PROJECT REPORT

1. Revenue Analysis of Products

Identify the products generating the highest and lowest revenue.

The query calculates total revenue per product and extracts both the highest and lowest revenue-generating products.

Key Insights:
- Revenue is concentrated in a few high-performing products.
- Low-performing products contribute minimal financial value.

Strategic Recommendation:
- Maintain uninterrupted stock for top-performing products.
- Promote low-performing products through targeted campaigns.

2. Revenue by Product Line

Determine which product lines generate the highest revenue.

The query aggregates revenue by product line using SUM and GROUP BY.

Key Insights:
- Classic Cars dominate overall revenue.
- Some product lines contribute minimally.

Strategic Recommendation:
- Reinforce investment in high-performing lines.
- Optimize low-performing categories.

3. Top Customers and Purchasing Behavior

Identify top customers and analyze their purchasing patterns.

The query calculates total spending, order frequency, and average order value.

Key Insights:
- A small percentage of customers contribute major revenue.
- Purchasing behavior varies between bulk buyers and frequent restockers.

Strategic Recommendation:

- Assign dedicated account managers.
- Introduce loyalty incentives.

4. Seasonal Trends and Order Volume

Examine how order volume and revenue vary over time.

The query groups data by year and month to identify seasonal trends.

Key Insights:
- Revenue peaks during specific months.
- Some months show lower business activity.

Strategic Recommendation:
- Increase inventory before peak seasons.
- Launch promotions during slow periods.

5. Order Status Impact

Analyze how different order statuses affect revenue.

The query groups orders by status and calculates total revenue.

Key Insights:
- Majority revenue comes from shipped orders.
- Cancelled or disputed orders cause financial leakage.

Strategic Recommendation:
- Reduce disputes and cancellations.
- Improve processing efficiency.

6. Employee Contribution Analysis

Evaluate employee performance in revenue generation.

The query calculates revenue handled per sales representative.

Key Insights:
- Some employees manage high-value accounts.
- Performance variation exists.

Strategic Recommendation:

- Introduce performance-based incentives.
- Implement mentorship programs.

7. Credit Limit Impact

Determine whether customer credit limits influence purchasing behavior.

The query compares total order value with credit limits.

Key Insights:
- High utilization customers may face restrictions.
- Some customers underutilize their limits.

Strategic Recommendation:
- Consider credit extensions for reliable customers.
- Reassess inactive high-limit accounts.

8. Revenue by Country

Identify countries contributing the highest revenue.

The query aggregates revenue and customer count by country.

Key Insights:
- Revenue is concentrated in specific regions.
- Some regions show growth potential.

Strategic Recommendation:
- Increase marketing in strong regions.
- Explore expansion opportunities.

9. Average Time to Ship

Measure shipping efficiency across product lines.

The query calculates the average difference between order and shipped dates.

Key Insights:
- Some categories experience longer delays.
- Delay patterns indicate warehouse inefficiencies.

Strategic Recommendation:

- Optimize warehouse layout.
- Standardize picking and packing procedures.

10. Products with Zero Sales

Identify products that have never been sold.

The LEFT JOIN identifies products without order records.

Key Insights:
- Unsold products represent locked capital.
- Indicates mismatch between supply and demand.

Strategic Recommendation:
- Conduct clearance campaigns.
- Review procurement strategy.

Executive Summary

This project converts raw transactional data into actionable business insights. It highlights revenue concentration, customer dependency, operational bottlenecks, and inventory inefficiencies to support data-driven decision-making.

Final Conclusion

This SQL mini project demonstrates practical application of relational database concepts, aggregation techniques, join operations, and analytical reasoning. It shows how structured query analysis supports business optimization.